

ACCELERATING AMERICA'S FOOD WASTE REDUCTION: A THREE-PILLAR
FEDERAL POLICY FRAMEWORK

GW Data Science Association Datathon 2026 | March 19, 2026 Team: Hritik Majgaonkar, Nidhi Naidu, Divya Mistry, Alexander D. Silberman

THE PROBLEM

The United States generates approximately 70 million tons of food surplus every year. Based on 15 years of data from ReFED's Food Waste Monitor (2010–2024), 71.5% of this surplus - 727.7 million tons cumulatively - is preventable. During this same period, 1,652 billion meals were wasted, \$2.4 trillion in economic value was destroyed, and food decomposing in landfills generated 3.02 billion metric tons of greenhouse gas emissions. Meanwhile, millions of Americans experience food insecurity. The communities that need food most are surrounded by surplus that never reaches them.

The USDA and EPA set a national goal in 2015: reduce food waste 50% by 2030 from the 2016 baseline. We are now in 2026. Using machine learning to project each state's current trajectory forward, we found that zero out of 50 states are on track to meet this goal. Not one. Forty-seven states will miss the target. Two states are actively getting worse. At the current rate of improvement, the US will still waste 63 million tons annually by 2030 missing the 38 million ton goal by 25 million tons. This is not a state policy problem. It requires federal intervention.

WHAT THE DATA SHOWS

Analysis of the ReFED cause-sector matrix identifies three combinations responsible for the majority of preventable surplus:

Cause + Sector	Scale	Root Cause
Farm: produce left unharvested	231.6M tons	Price crashes make harvesting unprofitable; 63M tons is marketable food simply abandoned
Foodservice: plate waste and overproduction	153.5M tons / \$1.78 trillion	Plate waste alone is 119.8M tons the single largest specific cause in the dataset

Cause + Sector	Scale	Root Cause
Farm: produce left unharvested	231.6M tons	Price crashes make harvesting unprofitable; 63M tons is marketable food simply abandoned
Retail and residential: date label confusion	49.8M tons	Competing label formats cause consumers to discard safe food prematurely

Four machine learning algorithms (Random Forest, Gradient Boosting, Ridge Regression, ElasticNet) were compared using Leave-One-Out cross-validation on a 50-state dataset. Ridge Regression achieved the best performance, explaining 57% of variance in state food waste rates. Two independent algorithms agreed on the same top drivers: harvest loss rate (confirmed as a causal driver via PC algorithm causal discovery), foodservice economic intensity, and recovery infrastructure quality. Counterfactual simulation shows that raising all states to top-quartile performance on harvest losses and foodservice efficiency would prevent 8.2 million tons per year, nearly two-thirds of our policy's 13 million ton annual target.

One proof of concept already exists: Wisconsin's food surplus rate (3.1%) is less than half of what its structural characteristics would predict (7.4%). The difference is traceable to Wisconsin Act 287 (2013), one of the first mandatory organic waste diversion laws in the US. If all states matched Wisconsin's performance, national waste would fall 69%, exceeding the 2030 goal. We do not need to theorize what works. Wisconsin is already doing it.

THE RECOMMENDATION: THREE-PILLAR FEDERAL POLICY

Pillar 1 — The Farm Recovery Act *Target: 231.6 million tons of unharvested farm surplus*

Causal discovery confirmed farm harvest losses as a root cause not just a correlate of overall food waste. Reducing harvest losses to top-quartile levels nationally would prevent 3.4 million tons per year.

- **Federal gleaning mandate:** large commercial farms must permit community gleaning after primary harvest. Already piloted successfully in California and New York.
- **Crop insurance reform:** extend USDA coverage to "not harvested due to market price collapse" losses, removing the financial incentive to abandon crops.
- **Produce price floor contracts:** USDA-backed forward contracts guaranteeing minimum prices during market gluts.
- **Emergency harvest tax credit:** automatically activated when spot prices fall below a USDA threshold.

Cost: \$1.3 billion | Value recovered: \$30 billion | ROI: 24×

Pillar 2 — Foodservice Efficiency and Donation Standard *Target: 153.5 million tons of foodservice excess*

Plate waste is the single largest specific cause of food waste in the ReFED dataset. The counterfactual model shows reducing foodservice waste intensity to top-quartile levels nationally prevents 4.8 million tons per year, the highest single-intervention impact of any policy lever tested.

- **Mandatory surplus reporting:** chains with 50+ locations must track and report annual food surplus by category, creating accountability and a national baseline.
- **Demand forecasting tax credit:** federal tax credit for AI-based demand forecasting adoption, which pilots have shown reduces overproduction by 10-15%.
- **Good Samaritan Act expansion:** update existing law to explicitly cover app-mediated donations (Too Good To Go, MealConnect). Legal liability ambiguity currently deters donation; one regulatory clarification removes this barrier at zero cost.
- **Federal portion transparency standard:** require chain restaurants to display default portion sizes, reducing plate waste without reducing customer satisfaction.

Cost: \$3.4 billion | Value recovered: \$356 billion | ROI: 106×

Pillar 3 — Federal Date Label Standardization *Implement first Target: 49.8 million tons of retail and residential surplus*

The US currently uses more than three competing date label formats. Consumers cannot reliably distinguish quality dates ("best by") from safety dates ("use by"), so they discard safe food. The USDA and FDA proposed standardization in 2019. It never became law. We are reviving it with quantified backing.

- **Federal mandate for two labels only:** "Best If Used By" (quality) and "Use By" (safety-critical only). No new infrastructure required. One rulemaking.
- **Retailer clearance incentive:** federal tax deduction for retailers offering reduced-price near-expiry sections.
- **K-12 food literacy:** incorporate date label and food storage education into existing federal school nutrition program curricula.

Cost: \$0.03 billion | Value recovered: \$45 billion | ROI: 1,493×

This is the highest-ROI food policy intervention available to federal regulators. Because it is a regulatory change, only no subsidies, no infrastructure, no ongoing program, every dollar spent returns \$1,493 in recovered food value. It can begin under existing FDA and USDA authority in 2025, 18 months before the other pillars require legislation.

COMBINED IMPACT

A 20% reduction across all three target cause groups, phased in through legislation enacted in 2027 and implemented through 2029:

Outcome	Value
Food surplus prevented per year	87 million tons
Economic value recovered	\$431 billion
Meals redirected to food-insecure communities	21 billion meals
Greenhouse gas emissions avoided	32 million mtCO ₂ e
Total implementation cost	\$4.7 billion
Combined return on investment	93×

Honest note: this policy brings national waste to approximately 51 million tons by 2030, closing half the 25 million ton gap to the 38 million ton goal. The remaining gap requires the infrastructure and accountability systems this policy creates, extended with further ambition in subsequent legislative cycles. The alternative of doing nothing misses the goal by 25 million tons. These are not equivalent choices.

IMPLEMENTATION ROADMAP

Phase	Timeline	Action
1	2025–2026	Date label rulemaking (FDA/USDA joint) no legislation required
2	2026–2027	Farm Recovery Act: gleaning mandate + crop insurance reform + price floors

3	2027–2029	Foodservice reporting mandate + Good Samaritan expansion + forecasting credit
4	2030	USDA national review against 50% reduction goal

DATA AND METHODS

Primary data: ReFED Food Waste Monitor (2010–2024), five datasets. Supplementary: USDA ERS food insecurity rates, NOAA NCEI temperatures, USDA NASS produce output, US Census population - all fetched live from official government APIs. Machine learning: four-algorithm comparison with Leave-One-Out cross-validation; best model Ridge Regression (LOO $R^2 = 0.574$); SHAP feature attribution; PC algorithm causal discovery (Fisher-Z, $\alpha=0.10$); counterfactual policy simulation; Isolation Forest anomaly detection. Full code available in the submitted repository.

"The communities that need food most are surrounded by waste that never reaches them. This is not a resource problem — it is a policy problem. The data shows us exactly where to act."